

Sumit Jahagirdar

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AI architect and engineer, five years of experience. I build AI agents and the models behind them: deep learning across NLP, computer vision, and speech and signal processing; fine-tuned LLMs and VLMs; custom detection and embedding models. I ship 0-to-1 as the sole technical owner: scoping, architecture, implementation, and production deployment, including multilingual (Japanese) agents at enterprise scale.

EXPERIENCE

Lead Data Scientist — Aays / Aidetic, Bengaluru

Jun 2022 – Present

Technical architect, AI Center of Excellence. Responsibilities: technical presales, solution architecture, client scoping, and leading a small engineering pod through delivery.

- **Agent Factory, Docomo Group.** Built and deployed a Databricks-native Japanese multilingual data agent serving 400 users across 166+ tables and 1,100+ columns of governed data. Responsible for backend, UX flow, APIs, and production deployment.
 - Wrote the orchestration, error handling, and processing layers myself, without an agent framework. The primitives are simple and configurable enough for an analyst to maintain the system.
 - State-separated runtime isolating conversation, execution, semantic, and data-object state. A Python engine computes statistical models on governed data at query time. Stateful memory, deterministic lineage, ACL-aware tool access, metadata registries, and evaluation hooks make responses reproducible and auditable.
- **Document extraction for private equity, 73 Strings.** Trained and deployed multimodal extraction models (LayoutLM and RCNN encoders, custom object detection for chart elements, chart-to-table reconstruction); reduced analyst effort in standardizing fund financials by ~90%. Parses 200+ page credit agreements into relational databases via segmentation, cross-reference tracing, and clause normalization.
- **Trade Fund Manager, P&G ANZ and Japan.** Chatbot for sales teams to query trade-fund data and ML models: generalized text-to-SQL with calibrated template tools and generative fallback, long and short-term memory, and secure data and tool access interfaces.
- **Speech AI, Mihup AI and LoginRadius.** High-throughput Whisper ASR pipeline with LLM call-QA grading; vLLM batching and prompt-redundancy reduction lowered latency and cost. Built a Descript-style video editor that compiles transcript edits into cuts, cloned speech, and lip-sync patches.
- **Applied ML.** Insurance damage classification for Acko with Databricks (YOLO plus LLM validation, MLflow/Delta continuous retraining). Fine-tuned CodeBERT and Llama for vulnerability detection in the DARPA AI Cyber Challenge, with automated evaluation harnesses (Panasonic/Aitomatic). Trained contrastive clause embeddings with few-shot semantic caching for search over 1,000+ legal classes (Cognizer).

Software Engineer, Edison AI — GE Healthcare, Bengaluru

Aug 2021 – May 2022

- Microservices and a unified Python SDK for clinical ML workflows on AWS Lambda and Kubernetes. Winner, GE Precision Health hackathon.

Machine Learning Research Intern — NVIDIA

Apr 2019 – Apr 2020

- ML research in medical imaging: few-shot learning and unsupervised clustering for pathology detection. Speaker, NVIDIA GTC 2020.

OPEN SOURCE

- **CodePulse:** MCP server that provides codebase context to Claude Code and Cursor. Typed semantic graphs with PageRank-ranked retrieval; reduces hallucination in AI-assisted coding.
- **Datalogist:** semantic inference engine that crawls warehouse partitions, clusters tables, and generates AI-ready data spaces and schemas.

SKILLS

AI core: deep learning (PyTorch, Transformers), NLP, computer vision (YOLO, OpenCV), speech and signal processing (Whisper/ASR, audio pipelines), fine-tuning (LLM/VLM/CNN, PEFT), contrastive learning, diffusion models, OCR, knowledge graphs

Agents & LLMs: AI agents, agent runtimes, RAG, MCP, evals, LangGraph/LangChain, OpenAI SDK, LiteLLM, vLLM, Ollama, prompt and cost optimization, text-to-SQL

Engineering: Python, FastAPI, Databricks/Spark, MLflow, Delta Lake, AWS (Lambda, SageMaker), Kubernetes, CI/CD, vector DBs

EDUCATION

B.Tech, Electronics and Communication Engineering — VIT Vellore, 2017–2021 (CGPA 8.03). Technical Head, IET VIT; Microsoft Student Ambassador (Top 30, APAC).